

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of M. Pharm (Quality Assurance)

University Theory Examination May 2015

PH816.08 Pharmaceutical Analysis - II

Date: 13.05.2015, Wednesday Time: 10.00 a.m. to 01.00 p.m. Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 A) Which are the important consideration for analysis of drugs from biological fluids? Why? 05
- B) Describe strategies and planning for bioanalytical method development. 05
- Q.2 A) Define bioanalytical method validation. Write objectives of bioanalytical method validation. Write definition, assessment methodology and criteria for (a) Specificity (b) Accuracy and (c) Stability studies. 10

OR

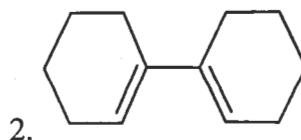
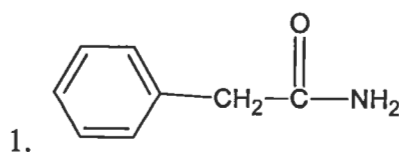
- Q.2 A) Discuss (a) Protein precipitation (b) Liquid-Liquid extraction and (c) Solid phase extraction techniques for preparation of samples for bioanalysis. 10
- Q.3 Write short notes on the following (Any three) 15
- (A) BA/BE study designs
- (B) Peptide mapping
- (C) Forensic analysis
- (D) Protein sequencing

SECTION-II

- Q.4 (A) Explain dissolution method development. 05
- (B) Describe in brief about isolation and characterization of impurities. 05
- Q.5 (A) Discuss application of chromatographic techniques in extraction and identification of plant constituents. 05
- (B) Explain the following. 1. Molecular ion, 2. Nitrogen rule, 3. Coupling constant, 4. Infrared spectroscopy 5. Impurities 05

OR

- Q.5 (A) Write a note on theories of diffusion. 05
- (B) Predict IR, NMR and MS data for the following compounds: 05



Q.6 **Answer the following (Any three)**

15

- (A) Forced degradation studies
- (B) Stability testing of phytopharmaceuticals
- (C) Chemical shift
- (D) Fragmentation Rule